Status	Finished
Started	Sunday, 11 October 2026, 3:02 PM
Completed	Sunday, 11 October 2026, 3:07 PM
Duration	5 mins 2 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned
b. The first element is returned
c. An error is raised
d. The search is considered unsuccessful ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first
b. By using a hash function
c. By dividing the list into halves
d. By comparing each element in the list sequentially ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Binary search
b. Linear search ✓
c. Interpolation search
d. Hash search

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Hashing
- b. Binary search
- c. Linear search ✓
- d. Dynamic search

Question **5**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. log n
- b. n ✓
- c. 1
- d. n/2

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search is more efficient than binary search on large lists.
- b. Linear search divides the list into halves.
- c. Linear search can be applied to both sorted and unsorted lists. ✓
- d. Linear search requires the list to be sorted.

Question **7**

Incorrect

Mark 0.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present ✗

Number values should only be present

b. There should be direct access to the middle element in any sublist

c. There must be a mechanism to delete and/or insert elements in the list

d. The list must be sorted

Question **8**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94

b. 90 and 94

c. 90 and 99 ✓

d. 89 and 99

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)

b. O👎

c. O(log n) ✓

d. O(n log n)

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues from the beginning
c. The list is sorted
d. The search continues in the left or right sublist ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By dividing the list length by two ✓
c. By comparing each element sequentially
d. By using a hash function

Question **12**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements
b. When the element is found in the list ✓
c. When the search algorithm finishes
d. When the list is sorted

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓
b. When the list is very small
c. When the list is unsorted
d. When the list contains duplicate elements

Question **14**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. At the beginning of the list
- b. In the left sublist ✓
- c. In the entire list
- d. In the right sublist

Question **15**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Determining whether an element is present in a list ✓
- b. Inserting elements into a list
- c. Deleting elements from a list
- d. Sorting elements in a list